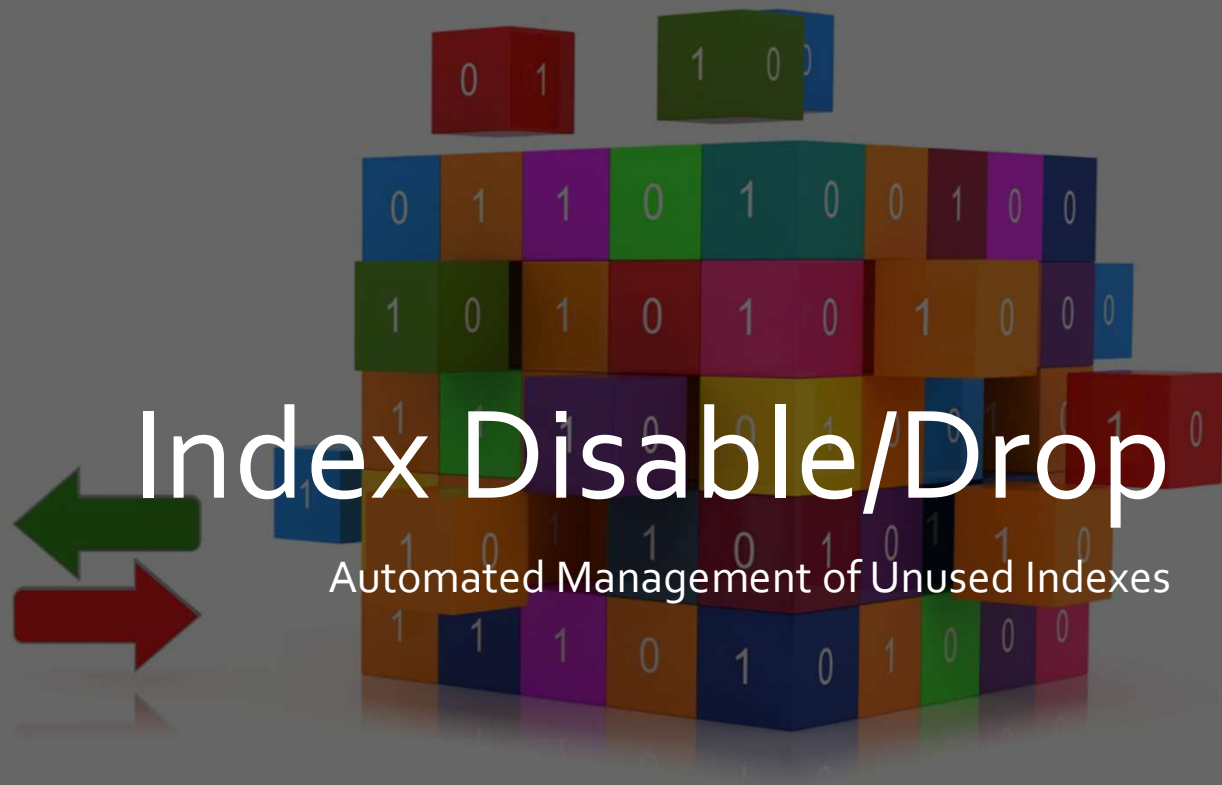




Index Disable/Drop

Automated Management of Unused Indexes

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What We Will Be Covering Today

- Developing an understanding of the impact of index growth on SQL Server performance and storage.
- Learning how to identify and manage unused indexes effectively.
- Exploring and understanding the automated processes for index disabling and dropping.
- Familiarizing with the supporting tables, stored procedures, and agent jobs.
- Gaining confidence in safely maintaining indexes to optimize database operations.

Understanding SQL Server Index Recommendations

- SQL Server often recommends multiple indexes based on query patterns.
- Not all recommended indexes improve performance equally.
- Blindly implementing all recommendations can cause overhead.
- Testing indexes in a controlled environment is essential.
- Careful evaluation helps avoid unnecessary resource consumption.



The Hidden Costs of Unused Indexes

- Unused indexes consume valuable disk space unnecessarily.
- Index maintenance can waste CPU and I/O resources.
- Memory is consumed by indexes even if they are rarely used.
- Write operations slow down due to index updates during data modifications.
- Regularly review and remove unused indexes to optimize performance.

Overcoming Fear of Dropping Indexes

- Understand the risks versus benefits of index removal.
- Use data-driven insights to guide your decisions.
- Leverage automated tools to monitor index usage.
- Implement safety checks before disabling indexes.
- Communicate changes clearly with stakeholders.





Automated Index Management: How It Works

- Nightly jobs track index usage and identify unused indexes over 120 days.
- Potentially unused indexes are disabled after safety checks and testing.
- Notifications alert stakeholders before indexes are dropped.
- The final index drop occurs on a specified date, maintaining records for review.
- Automation reduces manual effort and minimizes performance risks.



Key Tables in Automated Index Management

- `idxdd.IndexesToDrop` records index change history and scripts for management of disabled indexes. It stores scripts to create, rebuild, or drop the indexes selected from the `IndexesToDisable` table.
- `idxdd.IndexesToDisable` stores usage info on indexes which are eligible to be disabled. These are emailed to the DBAs for evaluation.
- `lkup.IndexUsage` stores daily index usage data, maintained for six months. It is used to gather only those indexes which have been unused in the last 120 days.
- `lkup.VendorDBs` prevents inadvertent drops on third-party database indexes. It is used to eliminate indexes from being recorded to the `IndexesToDisable` table, so they are never eligible to disable/drop.
- These tables enable precise tracking and management of indexes.

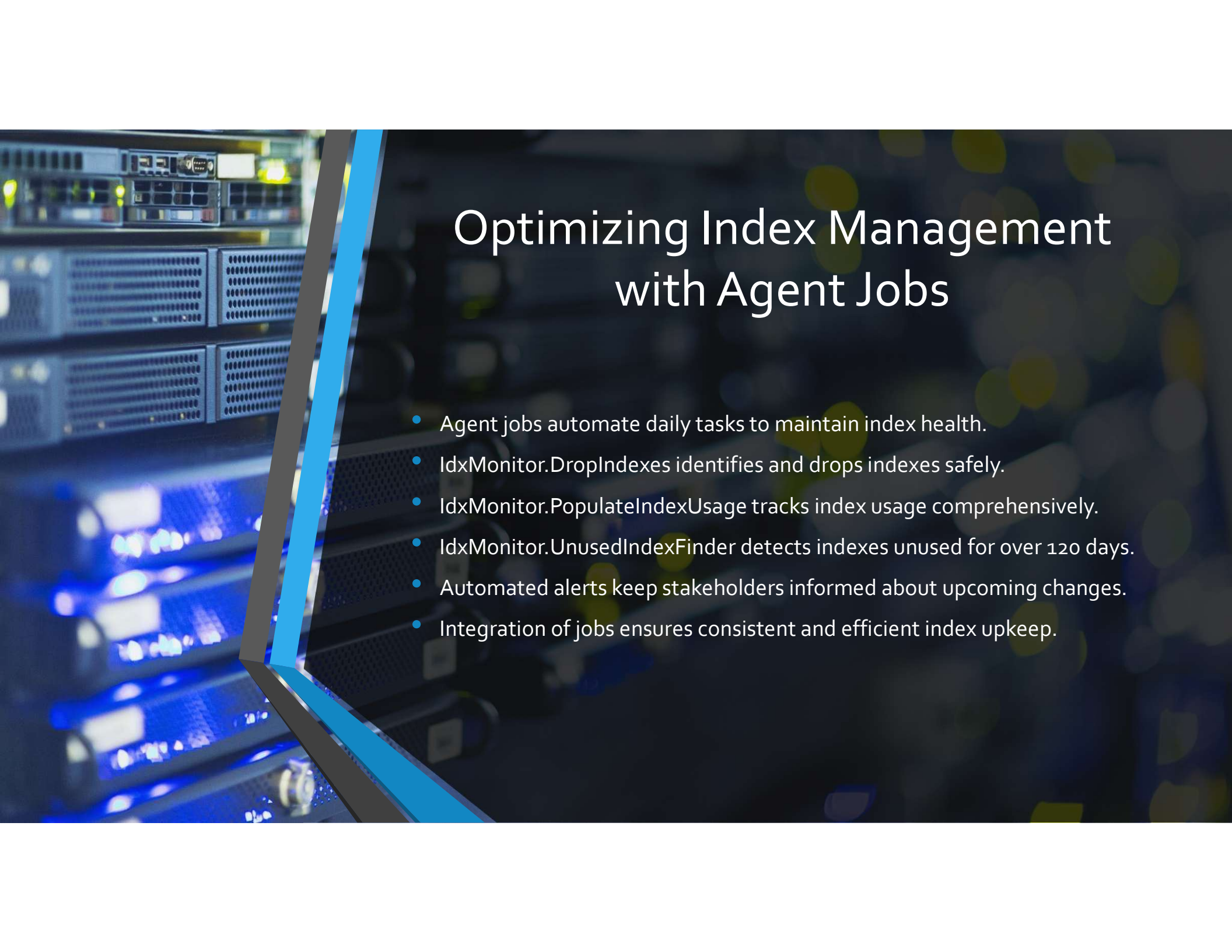


Enhancing Stored Procedures for Efficient Index Management

- Stored procedures automate critical index management tasks efficiently.
- Notification procedures alert stakeholders before index drops occur.
- Safety checks prevent unintended index disabling and dropping.
- Rebuild and usage tracking procedures maintain optimal index health.
- Integration with agent jobs ensures timely and consistent operations.

Stored Procedures Overview for Index Management

- idxdd.EmailIndexesToDrop sends notifications to change submitters daily for five days before indexes are dropped.
- idxdd. EmailUnusedIndexes alerts for indexes unused for 120 days.
- idxdd. FindIndexesToDisable identifies indexes eligible for disabling.
- idxdd. IndexDisable safely disables indexes after safety checks.
- idxdd. IndexDrop removes indexes on specified dates.
- idxdd. IndexRebuild restores disabled indexes when needed.
- lkup.PopulateIndexUsage records historical index usage data.



Optimizing Index Management with Agent Jobs

- Agent jobs automate daily tasks to maintain index health.
- `IdxMonitor.DropIndexes` identifies and drops indexes safely.
- `IdxMonitor.PopulateIndexUsage` tracks index usage comprehensively.
- `IdxMonitor.UnusedIndexFinder` detects indexes unused for over 120 days.
- Automated alerts keep stakeholders informed about upcoming changes.
- Integration of jobs ensures consistent and efficient index upkeep.



Questions?

Let's see it for ourselves.